INCH-POUND
MIL-DTL-32015/4(AS)
6 May 2004

DETAIL SPECIFICATION SHEET

MICROPHONES, LINEAR, MBU-12, MBU-20 AND MBU-23 OXYGEN MASK

This specification is approved for use by the Naval Air Systems Command, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32015.

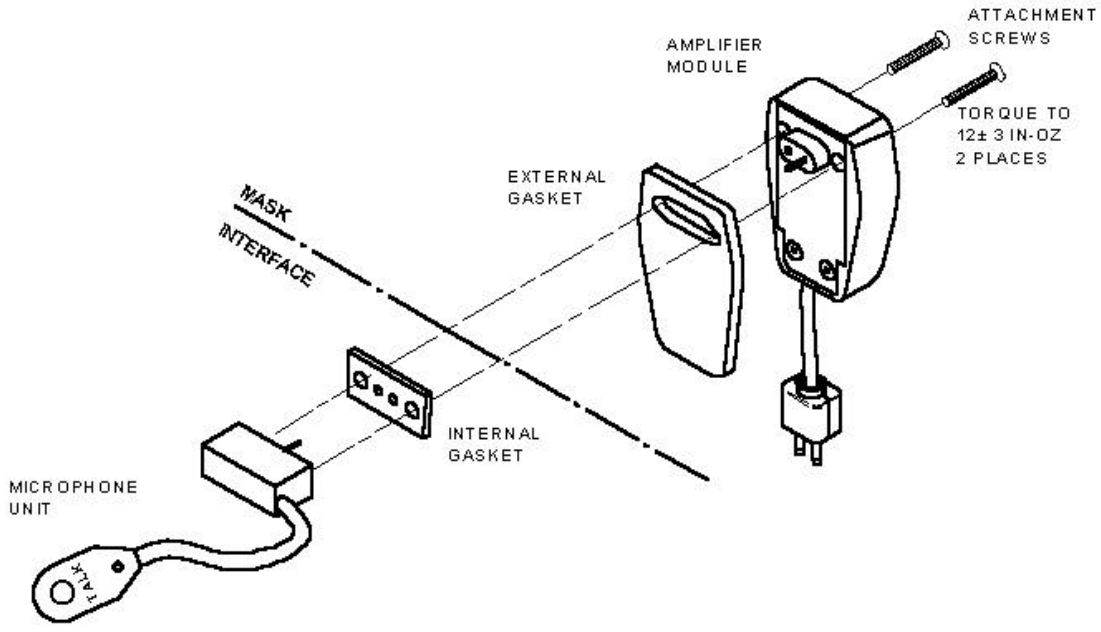



FIGURE 1. Amplifier w/Microphone assembly, M32015/4-1 (exploded view)

AMSC N/A FSC 5965

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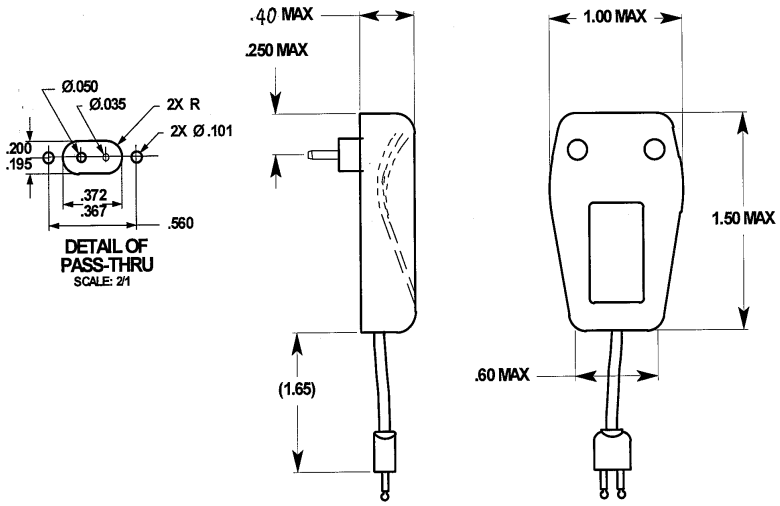



FIGURE 2. Amplifier assembly.

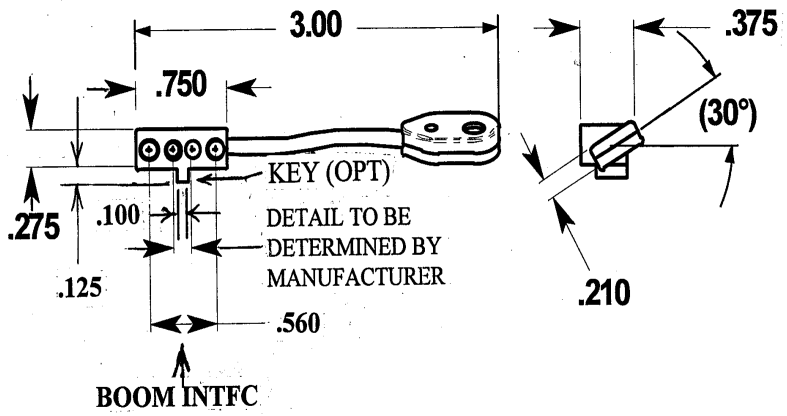



FIGURE 3. Microphone unit



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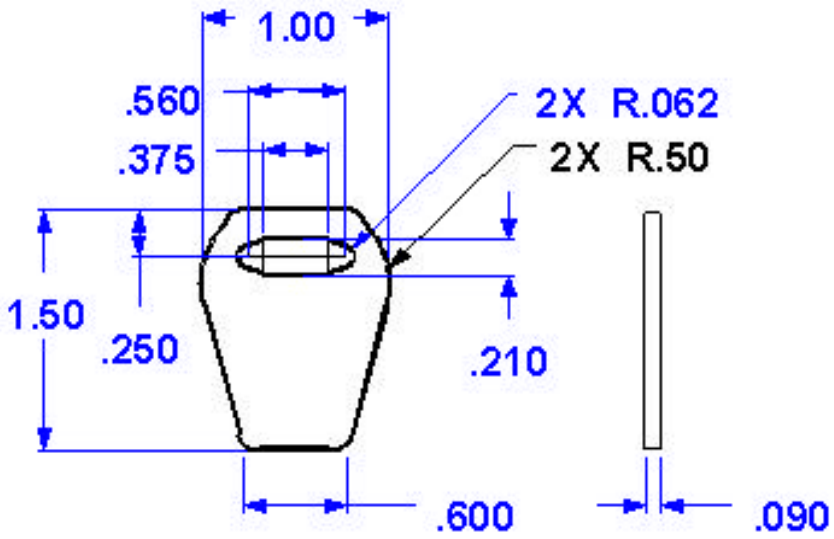



FIGURE 4a. External gasket.

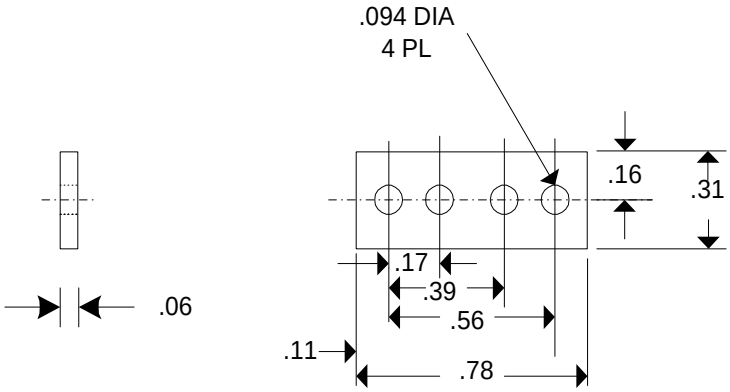



FIGURE 4b. Internal gasket.

NOTES

1. All dimensions in figures 4a and 4b are nominal except where indicated. Actual gasket dimensions shall be consistent with amplifier case design.
2. Material: Silicone Rubber, commercial grade.



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APPLICABLE DOCUMENTS

| | |
|---|---|
| NASM35275 | Screw, Machine-Drilled Fillister Head, Slotted, Corrosion Resisting Steel, Passivated, UNC-2A |

(Application for copies of NASC publications should be addressed to the Aerospace Industries Association, 1250 Eye Street NW, Washington DC 20005, or matthew@aia-aerospace.org.)

APPLICABLE DRAWINGS

AIR FORCE

| | |
|---|---|
| 57B12662 | Connector, Plug, Electrical U-173/U |
| 57B13664 | Tip-Electrical, Plug Connector |

(Copies of the above drawings are available from Defense Supply Center Columbus, DSCC-VTCD, P.O. Box 3990, Columbus, OH 43216-500, or www.charles,mcgrath@dla.mil.)

REQUIREMENTS:

DESIGN and CONSTRUCTION: See figures 1 through 4.

MICROPHONE/ AMPLIFIER INTERFACE: The microphone amplifier interface at the oxygen mask surface shall be sealed to protect the electrical contacts from the severe moisture effects inside the oxygen mask. The external and internal gaskets (see figures 4a and 4b) shall be permanently attached to the amplifier and microphone unit respectively (see figure 1).

CONFIGURATION: The microphone assembly is comprised of an amplifier assembly and a microphone assembly. The microphone assembly shall contain the microphone element, and only the electronics necessary for transfer of the audio signal to the amplifier. To the maximum extent practical, all electronics shall be contained within the amplifier assembly rather than the microphone element assembly to minimize the risk of a fire potential within the oxygen mask. Voltages and currents in the microphone assembly shall not create sparks or any other adverse effect, which presents a fire potential inside the oxygen mask.



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COMPONENT PARTS: The microphone assembly shall be supplied with the following:

1 Amplifier assembly (see figure 2).
1 Microphone unit (see figure 3).
1 Gasket (see figure 4a) which shall be attached to the bottom of the amplifier via adhesive.
1 Gasket (see figure 4b) which shall be attached to the microphone boom interface via adhesive.
2 NASM35275-210 machine screws, fillister head (2-56 x 3/4 lg.). Alternate mounting hardware may be used if acceptable to the procuring activity.

PART or IDENTIFYING NUMBERS (PIN): M32015/4-1

WEIGHT: 62-g max. for the amplifier w/microphone.

COLOR: LIGHT GRAY

CONNECTOR:

OUTPUT PINS: The output terminals shall contain two pins made in accordance with Air Force Drawing 57B12664. The pins shall be arranged as detailed in Air Force Drawing 57B12662 and together shall comprise the U-173/U connector.

AMPLIFIER TO MICROPHONE PINS: The pin configuration is not specified since the pins are within the system design and not required to be compatible with external connectors. The connector design shall be within the envelope dimensions shown for compatibility with the oxygen mask design.

AMPLIFIER AND MICROPHONE SUBASSEMBLIES: The amplifier and microphone subassemblies shall be constructed so, when assembled on the oxygen mask with the NASM 35275-210 screws (or alternate hardware suitable for the application), there is no damage to the amplifier/microphone assembly when tested in accordance with the pin retention and mounting torque requirements of the general specification. Alternate hardware may used for mounting as long as: it is compatible with in-service existing masks; it meets the requirements of the specifications; and hardware details are provided to the procuring activity for approval.

INPUT CURRENT: 8 ma max. at 10 VDC when connected as shown on figure 5.
NOTE: All tests except overvoltage and reverse polarity shall use this test setup.

IMPEDANCE: The electrical impedance of the microphone assembly shall be such that the unit shall meet the sensitivity and frequency response to a close sound source while connected to a 150 ohm ±1 percent load impedance.



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VOLTAGE OPERATING RANGE: 8 VDC to 12 VDC.

SENSITIVITY: 18mV (min) to 36mv (max) at 94 dB SPL with 10 VDC supply voltage using the normal test circuit on figure 5.

SENSITIVITY AT REDUCED BIAS VOLTAGE: Shall not vary more than 3 dB at 94 dB SPL from the 10 VDC established value when the supply voltage is varied between 8 and 12 volts.

SENSITIVITY AT ALTITUDE: ±3 dB of sensitivity at ground level while at a simulated altitude of 15,000 ft above sea level.

FREQUENCY RESPONSE AT GROUND LEVEL: The frequency response characteristic of the microphone, with moisture barrier in place, between 400 and 6,000 Hz, shall be within the response envelope shown on figure 6 herein.

FREQUENCY RESPONSE AT ALTITUDE: Within ± 5 dB of the frequency response at sea level while at a simulated altitude of 15,000 feet above sea level.

NOISE IMMUNITY: The noise-immunity characteristic of the microphone shall intersect the zero axis at 2,500 Hz minimum, and have a slope of at least 6 dB per octave (see figure 7).

LINEARITY: As specified in the general specification. The resistive load shall be 150 ±1 percent ohms.

HARMONIC DISTORTION: As specified in the general specification.

GAIN STABILIZATION: Within 100 ms when supply voltage is varied between 8 and 12 VDC.

OVERVOLTAGE PROTECTION: 30 minutes operation using the overvoltage test circuit shown on figure 8 with +28VDC applied

REVERSE POLARITY PROTECTION. No damage to microphone or preamplifier when subjected to -28VDC for 30 minutes using the overvoltage test circuit shown on figure 8.

EXPLOSIVE ENVIRONMENT: As specified in the general specification.

EXPLOSIVE DECOMPRESSION: As specified in the general specification.

ENVIRONMENTAL STRESS SCREENING (ESS): 24 hours of temperature cycling between -20 and +85 °C, with each cycle consisting of 1 hour at each temperature and the change between temperature occurring at a minimum rate of 5 °C per minute. The microphone assemblies shall be functionally tested prior to each cold to hot transition and upon completion of the temperature cycling.

MASK PRESSURIZATION: As specified in the general specification.



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CONDENSING HUMIDITY: As specified in the general specification.

MARKING:

M32015/4-1 **(10VDC)** (V-22 PROGRAM ONLY)*
MANUFACTURER'S NAME OR CAGE CODE
CONTRACT NUMBER
DATE CODE
SERIAL NUMBER
*The voltage designator shall be molded or engraved and ink-filled.

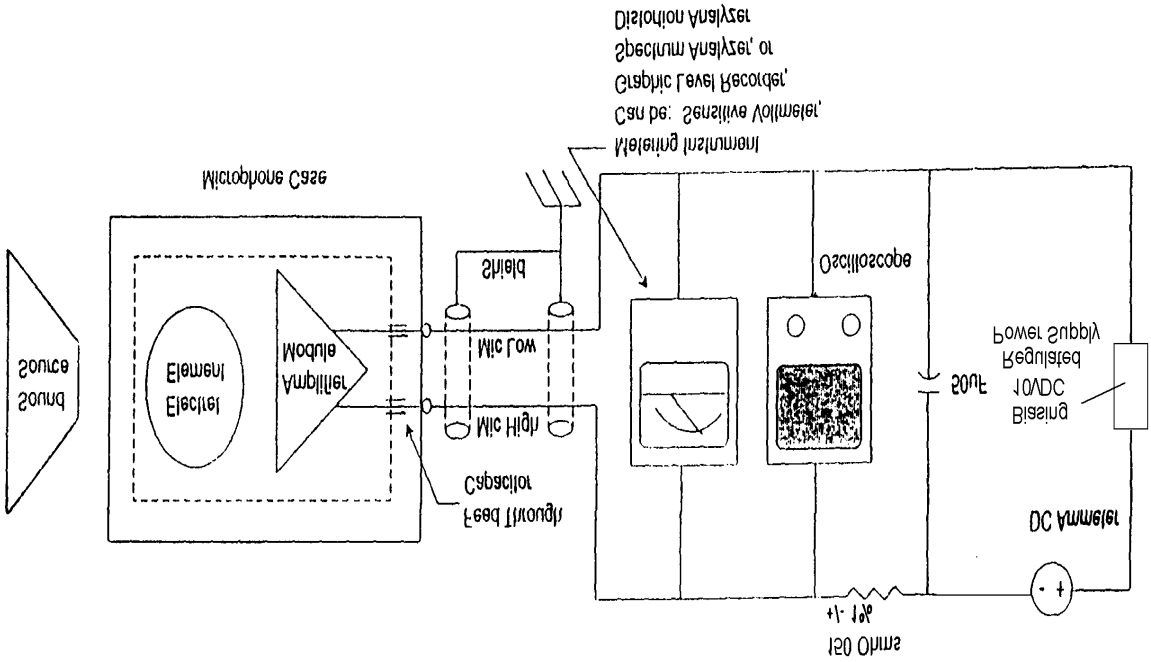



FIGURE 5. Test circuit schematic.
(for all tests except overvoltage and reverse polarity)



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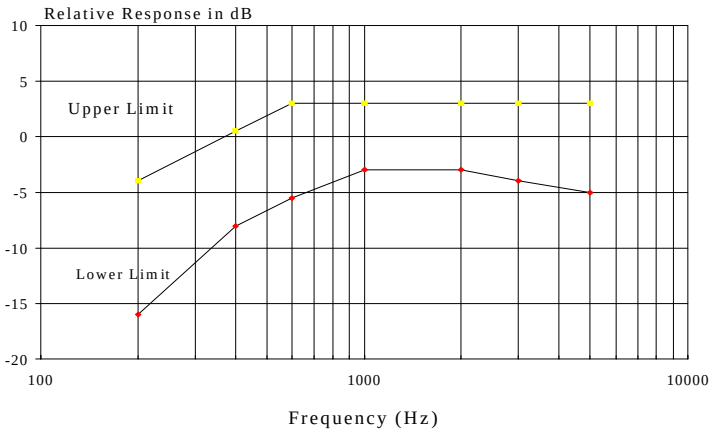



FIGURE 6. Frequency response limit curves.



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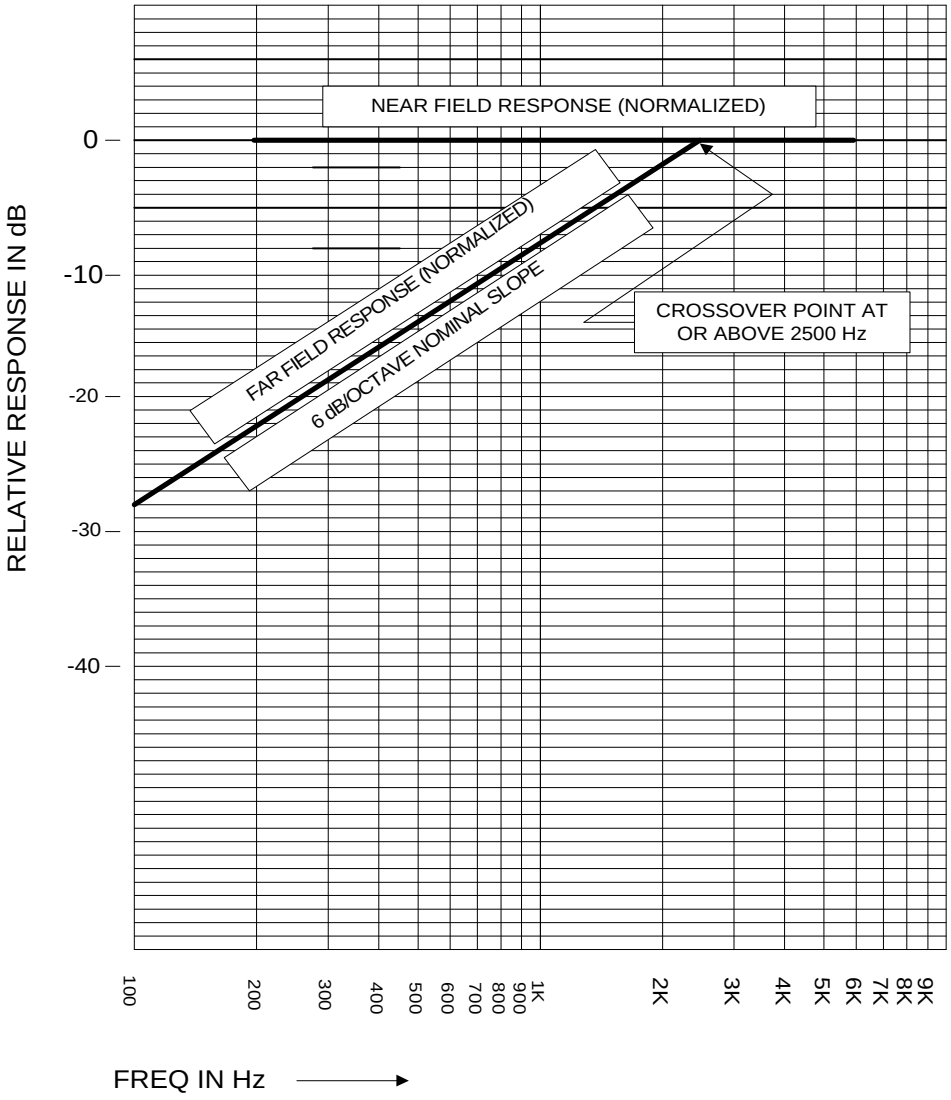



FIGURE 7. Noise immunity characteristics.



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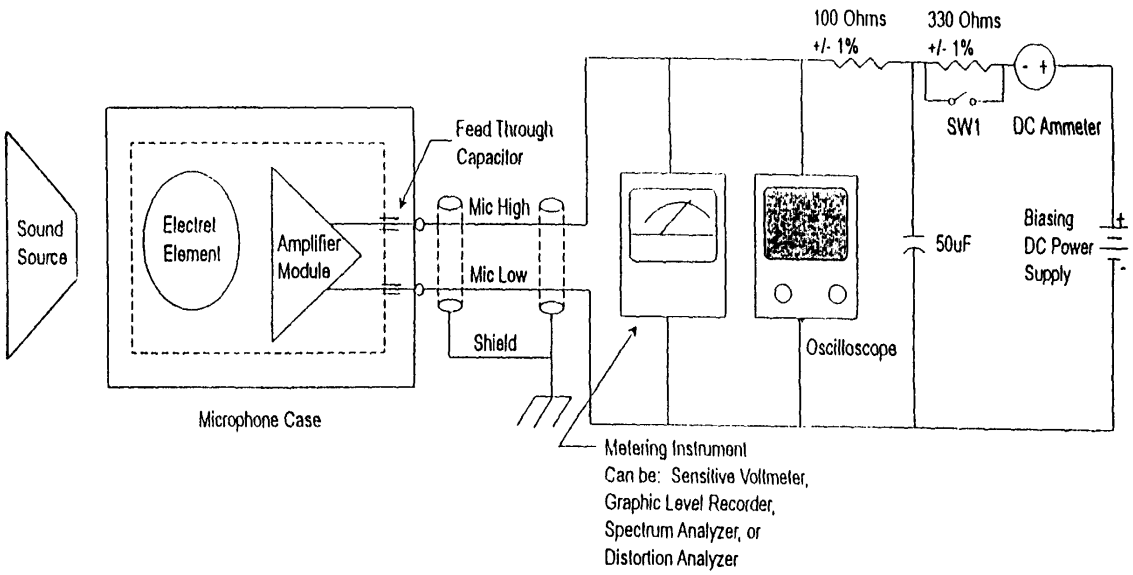



FIGURE 8. Overvoltage test circuit schematic.
(for overvoltage tests only)

INTENDED USE: The microphone, M32015/4-1, is a noise-canceling, DC-powered, high-gain microphone designed for use in a pressure-type oxygen mask. The microphone is intended for use as a part of the airborne communication system which will provide communication of high intelligibility under the extreme noise conditions encountered in military aircraft. The microphone is designed to work with 10 VDC biased intercommunication systems

Preparing activity:
Navy - AS
(Project 5965-0396)

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